

INTRODUCTION TO PYTHON PROGRAMMING



LESSON PLAN AND MATERIALS

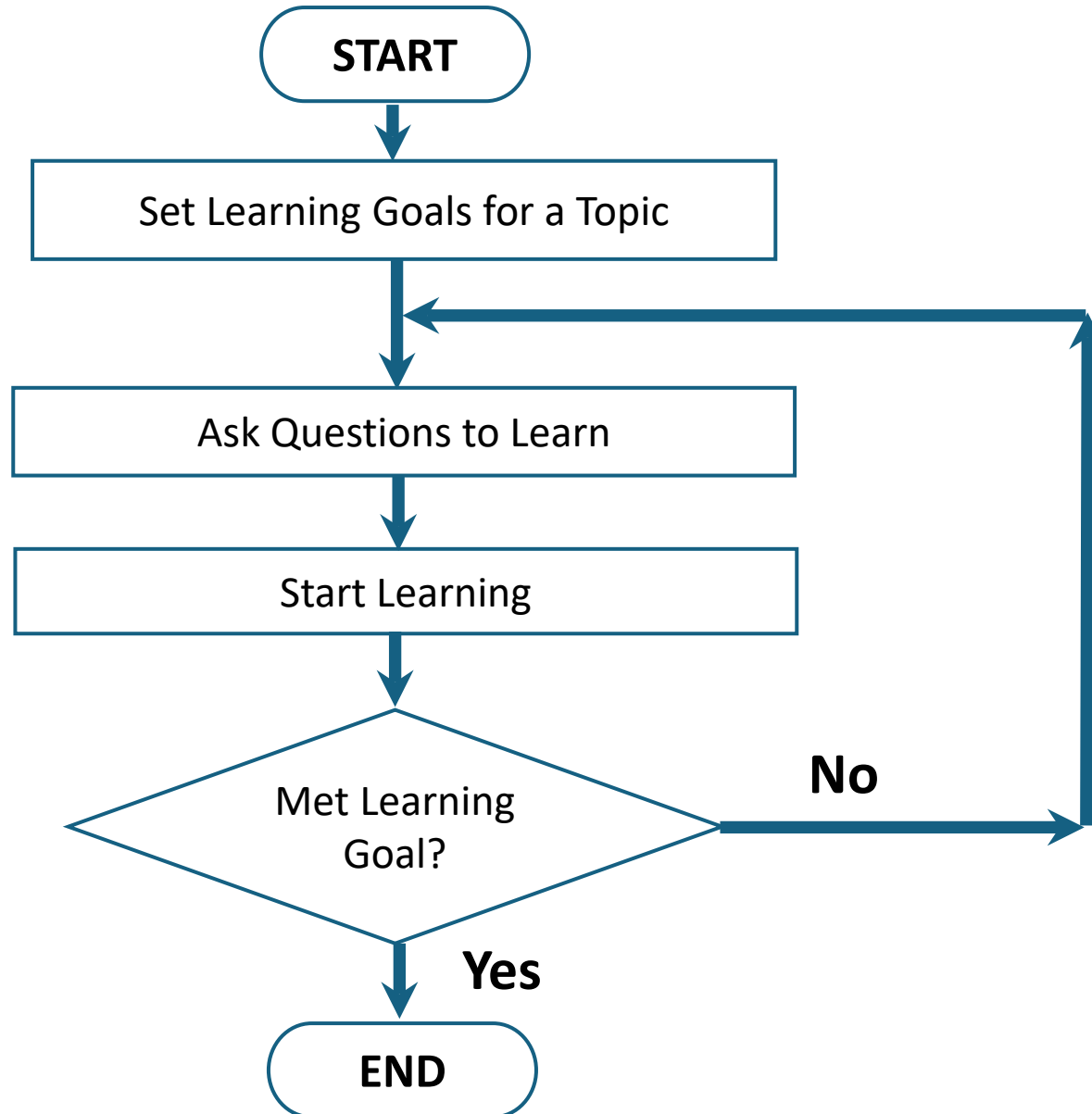
Understand the following

- Where to find lesson Plan and how to interpret them
- How to access the materials
- When are all the tests
- Student need to go through eLearning lecture on their own before coming for Lab

Extra **Free** online resource

- Python Essentials 1 (from Python Institute) : <https://edube.org/study/pe1>
- Sign Up : <https://edube.org/login>
- Our materials coverage is aligned to this online course.
- At the end of the semester, student may consider to take up their certification :
PCEP – Certified Entry-Level Python Programmer
 - Those who pass the certification, there are funding support to reimburse the cost of certification

LEARNING APPROACH



SET LEARNING GOALS

S-M-A-R-T goals stand for goals that are:

- Specific
- Measurable
- Attainable/ Achievable
- Realistic
- Time-bound

<https://rise.articulate.com/share/PyjgHOJ6kbYPKOH5CvXiuEw97sD72xWL>

SET LEARNING GOALS

Learning goal for this topic **Introduction to Python Programming.**

By the end of this practical, I want to be able to use python to

- Display some message to user
- Get some input from user like name, Year of Birth
- Ask user to enter Year of birth, calculate the age of this person and display it on screen

Specific : 3 tasks to fulfil

Measurable: Able to determine whether I have achieved it

Achievable: At my level, it is simple enough for me

Realistic: After some practice in Lab, it should be feasible

Time bound: Need to achieve it by end of the practical session

ASK GOOD QUESTIONS

What is KWL

Do you know there can be a structured way to direct your learning? You can use a KWL template!

KWL is a graphic organiser with three columns that can be used during the beginning, middle, and ending of a lesson. The 'KWL' stands for :

- What you **Know**
- What you **Want** to learn
- What you have **Learned** from the lesson/ material/ experience.

<https://rise.articulate.com/share/rPoNzrTZ4DnVDoevhTN7mugeg6kbND3a#/>

ASK GOOD QUESTIONS

What you **Know**:

- Visual Studio Code is the IDE that I use to write the python codes
- Python is a programming language that I need to learn

What you **Want** to Learn:

- How to use Visual Studio Code to create and run a program
- How to use python to display text message on screen
- How to use python to get inputs from user like name, Year of birth
- How to use python to calculate age of user from their Year of Birth and display it on the screen

WHAT IS A FUNCTION

Find the eLearning Topic **Input, Processing & Output**

Read through the following section

- What is a function in python



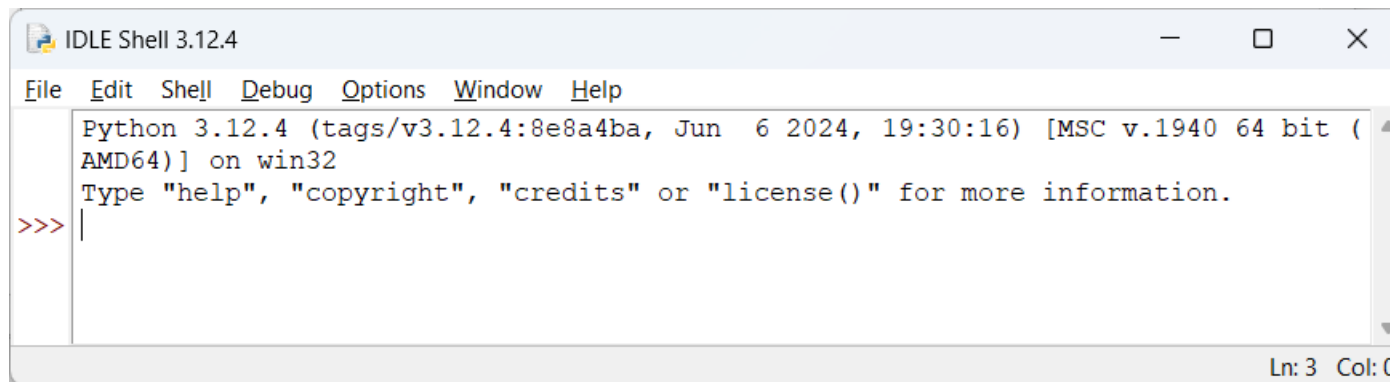
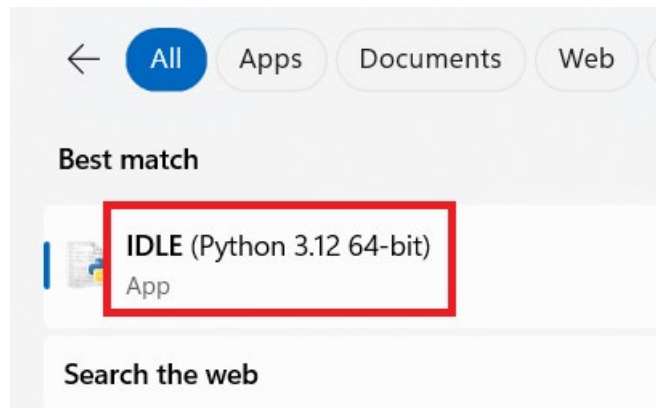
10 mins

We shall proceed on with Lab after this

PRACTICAL 1

Interactive Mode using Python Shell

- Run IDLE to start the python shell.
- We can use this to run simple short python codes



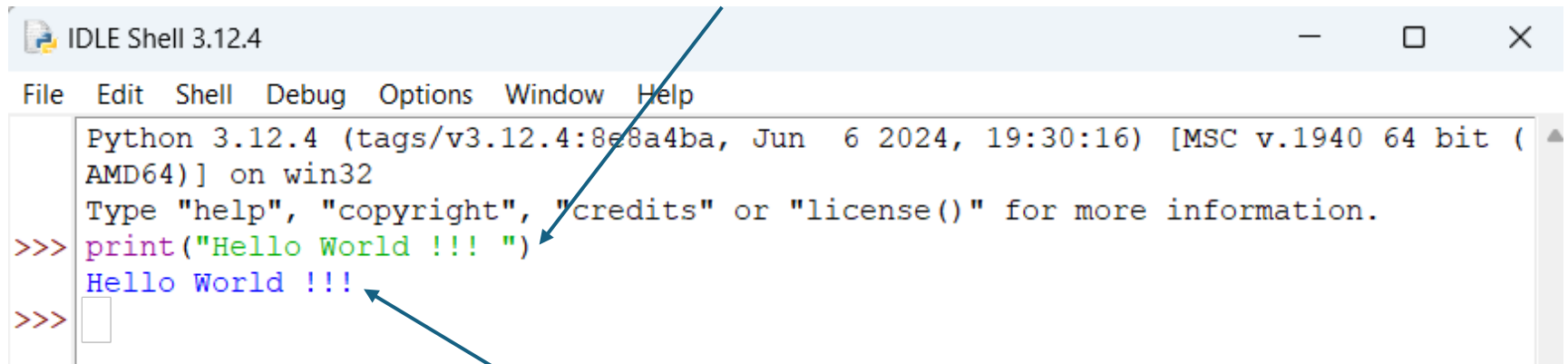
Our first python code

- The **print()** function

- We use the **print()** function to display message or text on the screen.
- We need to put the message for display in between the round bracket **()**
 - Eg. **print("Hello World")**
- The value we put inside the **()** is known as the **argument** of the **print()** function

We shall now print the message **Hello World !!!**

- At the IDLE, type the below and hit Enter key
 - **print("Hello World !!! ")**



The screenshot shows the IDLE Shell 3.12.4 window. The title bar reads "IDLE Shell 3.12.4". The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell window displays the following text:

```
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> print("Hello World !!! ")
Hello World !!!
>>>
```

Two blue arrows point from the text in the list above to the code in the shell. One arrow points from `print( "Hello World !!! " )` to the `print` function, and the other points from the same code to the output `Hello World !!!`.

You should see the text **Hello World !!!** being displayed.

- Note that the `" "` is not displayed. They are used to indicate the start and end of the text message to display

We shall now print the message **Hello World !!!**

- Instead of double quotes, try with single quotes
 - **print('Hello World !!! ')**

```
>>> print('Hello World !!! ')\n...\nHello World !!!\n>>>
```

We can also display the text message using **''**

- Python allows you to represent a text using either **''** or **"""**

Take note of the following

- Python code is case sensitive
 - **print()** is different from **Print()**
 - Try typing **Print**("Hello World !!! ")

```
>>> Print("Hello World !!! ")
Traceback (most recent call last):
  File "<pyshell#2>", line 1, in <module>
    Print("Hello World !!! ")
NameError: name 'Print' is not defined. Did you mean: 'print'?
>>> |
```

Python understand **print()** as it is a Built-in function but python cannot run **Print()**. Be careful to use the correct case.

- **NameError** means python cannot find a function **Print()**
- This is example of a **runtime error** where python is unable to locate and run the **Print()**

Take note of the following

- The correct brackets are important
 - **print()** is different from
 - print , print{ }, print[]
 - Try typing **print["Hello World !!!"]**

```
>>> print["Hello World !!!" ]
Traceback (most recent call last):
  File "<pyshell#4>", line 1, in <module>
    print["Hello World !!!" ]
TypeError: 'builtin_function_or_method' object is not subscriptable
>>>
```

Because you used the wrong bracket, python may treat it as a different command as square brackets [] are used for some other purpose

- This is another example of **runtime error**

Take note of the following

- Many codes must match open and closing characters. For example:
 - Round brackets ()
 - Double quotes " "
- Try typing the below without the matching close quotes
 - `print( "Hello World !!! )`

```
>>> print("Hello World !!! )  
...  
SyntaxError: unterminated string literal (detected at line 1)  
>>> |
```

When the matching double quote is missing, the message to display is considered unterminated (not ended properly).

- This is an example of **syntax error**

Exercise

Try to display the below Text **with the double quotes.**

```
Shelock exclaimed "Elementary my Dear Watson!!"
```


Exercise

Find the eLearning Topic **Input, Processing & Output**

Read through the following section

- Output with print()

Exercise

Try to display the below Text with **1 print() statement**

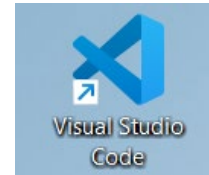
First Name	Last Name
Bruce	Banner
Steve	Rogers

Using Visual Studio Code

Create a project folder wk01 on Desktop

- We will put all our python files inside this folder

Start Visual Studio Code IDE



Set wk01 as the project folder

- **File – Open folder**
- Navigate to project folder : **wk01**
- Click on **Select Folder** to set the project folder

Create python file

- Click on **New File**
- Enter **Hello.py** (Note: Needs to end with **.py**)

Visual Studio Code

- Visual Studio Code is using scripted mode to run the program.
- It runs all the statements at once from file.

Running the program

- Type the below codes, then click on the run button



We shall see the output below at the **TERMINAL** tab



Does print() function returns any value?

Try typing and run the code below

Output

```
retValue = print("Hello World !!! ")  
print(retValue)
```

```
Hello World !!!  
None
```

None

- Python use a special value **None** to represent no value.
- Many of the function returns this value if it is not relevant or no result
- **print()** is only used to display message on screen. It is not relevant to return any value, so it returns **None**
- As you learn on the topic Decisions (**if**) later, you may use **None** for evaluation of results.

Comments

- They are used to tell you what the code does
- Especially when you are working in a group, your friend's comment helps you to understand the codes
- We use the **#** sign to inform python that anything that comes after the # sign are comments
- python will ignore these text and not interpret them
- See below code

```
1  # this is my simple Hello world program,  
2  print("Hello World !!! ")
```

Note:
comments are ignored

Output

```
Hello World !!!
```

Some common data types

str : string literals (Text data)

- The data is enclosed by **" "** or **' '** Eg. **"88"** is a string literal
- **+** : you can join multiple strings together using **+**. This is known as concatenation

```
print( "88" + "00" )
```

Output

```
8800
```

- ***** : you can multiply string with a number. The text duplicates and form a longer string

```
print( "81 " * 3 )
```

Output

```
81 81 81
```

Some common data types

int : Integer literal (whole number value)

- Numerical value without any “” or ‘’. Eg. 88 is an integer literal
- Normal math operations can be performed on it. Common operations includes
 - + : summation
 - - : subtraction
 - * : multiplication
 - / : division

```
print( 88 + 10 )
```

Output

98

float : float literal (floating points value)

- Same as integer except that it has decimal value. Eg. 88.0

Some common data types

Using type() function to display data types

```
var1 = "88"  
print(type(var1))
```

```
var2 = 88  
print(type(var2))
```

```
var3=88.0  
print(type(var3))
```

Output

```
<class 'str'>
```

```
<class 'int'>
```

```
<class 'float'>
```


Some common data types

Be mindful of the data types and **+** operations

```
a1 = "88"  
b1 = "10"  
result1 = a1 + b1  
print(" result1 =", result1 )
```

```
a2 = 88  
b2 = 10  
result2 = a2 + b2  
print(" result2 =", result2 )
```

```
a3 = "88"  
b3 = 10  
result3 = a3 + b3  
print(" result2 =", result3 )
```

Output

```
result1 = 8810  
(performs concatenation )
```

```
result2 = 98  
(performs summation )
```

```
Traceback (most recent call last):  
  File "c:\Temp\wk01\Hello.py", line 14, in <module>  
    result3 = a3 + b3  
              ~~~~~
```

TypeError: can only concatenate str (not "int") to str
(cannot perform string + integer !)

This is a runtime Error

Exercise (fix the error in this code)

There is a runtime error in this code using input()

```
age = input("Enter your age: ")
age = age + 1
print("You will be",age,"years old next year")
```

Find the eLearning Topic **Input, Processing & Output**

Read through the 2 sections and try to fix the error

- Input from Keyboard
- Reading numbers with input()

Exercise

Get user to enter name, current year, year of birth, height in cm.
Display the below message to user informing current age and height in m.

```
Enter your name: Roy  
Enter current year: 2025  
Enter year of birth: 2007  
Enter your height (cm): 174  
Hello Roy, you are 18 years old and 1.74 m tall.
```

Find the eLearning Topic **Input, Processing & Output**

Read through the following sections

- Formatted output with print()

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ASK GOOD QUESTIONS

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- How to use Visual Studio Code to create and run a program
- How to use python to display text message on screen
- How to use python to get inputs from user like name, Year of birth
- How to use python to calculate age of user from their Year of Birth and display it on the screen

What have you **Learnt** :

- Know how to create and run a simple program using Visual Code
- Know how to use **print()** statements to display text messages on screen
- Know how to use **input()** to get user inputs like name, year
- Done a program to get in name, age, current year, year of birth, calculates the age and display to user

LEARNING GOALS MET

Learning goal for this topic **Introduction to Python Programming.**

By the end of this practical, I want to be able to use python to



- Display some message to user



- Get some input from user like name, Year of Birth



- Ask user to enter Year of birth, calculate the age of this person and display it on screen

INSTALLATION AND SETUP

Student bring Laptop to Lab to install

- Refer to “Install_python_and_VSCode” document
- Install python
- Install Visual Studio Code
- Install python extension for Visual Studio Code

BYE

BYE